Lab 6

Question 1:

```
# ========================
# 1. Family Shoe Data
# ========================
family = ["Homer", "Marge", "Bart", "Lisa", "Maggie"]
shoe_color = {"Homer": "brown", "Marge": "blue", "Bart": "red", "Lisa": "yellow", "Maggie": "pink"}
shoe_size = {"Homer": 12, "Marge": 8, "Bart": 10, "Lisa": 7, "Maggie": 5}

print("\n--- Family Shoe Data ---")
for member in family:
    color = shoe_color[member]
    size = shoe_size[member]
    size_label = "large" if size >= 10 else ""
    print(f"{member} wears {size_label} {color} shoes size {size}".strip())
```

```
--- Family Shoe Data ---
Homer wears large brown shoes size 12
Marge wears  blue shoes size 8
Bart wears large red shoes size 10
Lisa wears  yellow shoes size 7
Maggie wears  pink shoes size 5
```

Question 2:

```
# ========================
# 2. Filter Even Numbers
# ========================
numbers = [386, 462, 47, 418, 233, 237, 412, 344]
print("\n--- Even Numbers (Stop at 237) ---")
for num in numbers:
    if num == 237:
        break
    if num % 2 == 0:
        print(num)
```

```
--- Even Numbers (Stop at 237) ---
386
462
418
```

Question 3:

```
# ========================
# 3. Sequential Arithmetic
# ========================
print("\n--- Sequential Arithmetic ---")
a = 5
print("Step 1:", a)
a += 6
print("Step 2:", a)
a *= 2
print("Step 3:", a)
a += 1
print("Step 4:", a)
a -= 9
print("Step 5:", a)
a /= 7
print("Step 6:", a)
```

```
--- Sequential Arithmetic ---
Step 1: 5
Step 2: 11
Step 3: 22
Step 4: 23
Step 5: 14
Step 6: 2.0
```

Question 4:

```
# ========================
# 4. Disk Space Monitor
# ========================
import psutil

usage = psutil.disk_usage('/')
percent_used = usage.percent
print("\n--- Disk Space Monitor ---")
print(f"Disk usage is at {percent_used}%")
if percent_used > 90:
    print("Warning: Disk usage is above 90%! Consider cleaning up some space.")
```

```
--- Disk Space Monitor ---
Disk usage is at 0.8%
```